Personalized Text Summarization Based on Gaze Patterns

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Abstract

With the explosive growth of online content, summarization has become important for users to grasp information quickly. However, existing summarization only provides static and one-size-fits-all results, e.g., a paragraph is summarized to the same short sentence, for different users, failing to satisfy users' diverse preferences. In our subjective study, such preference diversity on content summarization can be quite significant, e.g., the ground-truth summaries provided by different users on the same long text only have an average cosine similarity value of 0.383. This paper fills this gap by a personalized content summarization framework. First, we have collected a Chinese long text summarization dataset with gaze behavior of different users. Based on our measurement study on the dataset, we reveal that people' s gaze behavior and their preferred content summary have strong correlation, e.g., the preference of different parts of speech (POS); Second, we propose to incorporate such gaze-based "attention patterns" into text summarization, by designing a gaze-based key sentence extracting strategy, PRank (short for PersonalRank) which is then integrated with a conventional pointer generator model to satisfy different individuals. Our trace-driven experiments on the dataset verify the effectiveness of our design: our model outperforms baselines by at least 3 ROUGE-1 points.

Keywords-PRank; Gaze Patterns; Personalized Text Summarization

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1. Introduction

With the rapid development of the Internet, there is massive information flocking to us every day. Thus, how to help us to improve the efficiency of information acquisition becomes an attractive research topic. Particularly for textual information, automatic text summarization is the most widely used technology, which produces concise summaries to represent the original texts.

Generally, there are mainly three types of methods for text summarization: extractive models, abstractive methods, and the integration of them. Extractive methods [11] directly select several sentences from the original text by different metrics. In contrast, abstractive methods [12] can generate new keywords which may not exist in the original text but deliver a similar message to the readers. Existing RNN based abstractive models usually have issues on gradient vanishing or exploding when processing long texts. Recent studies [15] then propose hybrid extractive and abstractive models to handle long text issue: several sentences will be selected by the extraction module first, and then the new text consisting of the selected sentences will be regarded as the new input for the abstraction module to generate the final summary.

However, the methods above only provide static and one-size-fits-all results, falling to satisfy users' unique "tastes", because of their different backgrounds and

preferences. According to our subjective measurement studies, such uniqueness can be quite significant. To this end, we propose to generate personalized text summary by tackling the following challenges: How to obtain the preferences of users on short summaries, and how to incorporate such preferences into text summarization.

In this paper, we propose a personalized text summarization framework based on gaze patterns. In particular, we build a summary preference profile for each user based on their gaze patterns when reading content on the screen. Then we learn the correlation between the gaze patterns and their preferred summaries. Based on the correlation, personalized summaries are generated using the gaze profile guided sampled sentences. Our contributions can be summarized as follows.

First, we design subjective experiments to study the correlation between people's gaze patterns and their personalized summaries. We use eye-tracking devices[1] to collect the gaze traces from 50 people when they are reading 100 articles[2] and submit their own summaries (as ground truth) of the articles. Our measurement studies reveal the following insights. 1) People have quite different gaze patterns, even when they are reading the same article; 2) The gaze patterns are closely related to the ground-truth summaries they submit.

Second, based on the above insights, we propose PRank (short for PersonalRank) that builds individual's gaze patterns for content summarization. It consists of the following building blocks. First, PRank selects key sentences from the original article, not only according to their general importance in content but also the "matching" to a user's attention profile according to the gaze patterns. Second, using the key sentences selected, we use a pointer generator [13], which is pre-trained with a huge dataset[3] for the content abstraction. In this way, PRank takes both content and user preference into consideration. Be noted that our design can be extended to applications that cannot collect gaze patterns for everyone: a small fraction of gaze patterns from the group can be used to compose others' attention profiles.

Finally, according to the experimental results, the personalized summaries generated by our design outperforms baselines by up to 3 ROUGE-1 points.

The rest of the paper is organized as follows. We survey related work in Sec. 2. We study readers' gaze patterns and the correlation between gaze and their preferred content summaries in Sec. 3. The details of our design are in Sec. 4. We evaluate the effectiveness of our design in Sec. 5 and conclude the paper in Sec. 6.

[1] Tobii Eye Tracking 4C

[2] Netease News and Tencent News

[3] TTNews

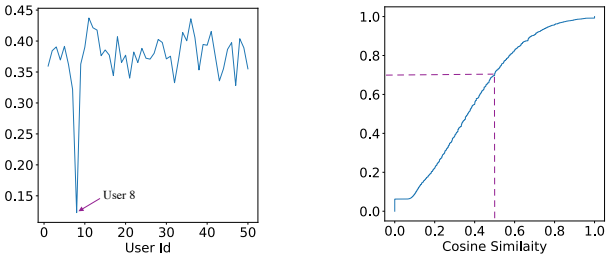
2 Related Work

Most of the previous studies focused on static text summarization rather than personalized text summarization. Neto et al. [11] used an extractive model to identify key sentences in the source document and reorganize them as a summary. And there are several key sentence extraction algorithms, such as LexRank [3], TextRank [6]. Because RNN based sequence-to-sequence model with attention has achieved great success in various natural language processing fields, Rush et al. [12] and Nallapati et al. [10] applied seq2seq model with attention to abstractive text summarization, and they perform well in generating short texts. See et al. [13] proposed a pointer generator model with coverage to solve the problem of out-of-vocabulary (OOV) words and duplicate words. Wang et al. [15] tried to integrate extractive and abstractive model for long text summarization.

For personalized content summarization, Diaz et al. [2] used a user model in the form of a vector of weighted keywords, while the users have to insert keywords and weights into the model manually. Campana et al. [1] propose to automatically build a user model according to the sentences recently read by the user, and the sentences which are the most similar to the representative sentences from the user model are reproduced as the summary. Moro et al. [9] proposed a method extracting important information from the document according to some sources of personalization, such as annotations.

In recent years, there have been already some studies to use gaze behavior in many different scenes. Some of them used gaze behavior to evaluate certain aspects of the text, such as readability [7] and grammaticality [4]. Mathias et al. [5] proposed to predict the rating of text quality using gaze behavior. Mishra et al. [8] use gaze behavior in cognition-cognizant sentiment with multi-task learning.

A limitation of previous studies for personalized content summarization is that they only infer users' preference from the content aspect. In this paper, we propose to explore the correlation between readers' gaze attention and the personalized content summarization, and generate summaries accordingly.



(a) (b)

Figure 1: (a) shows similarities between each user and others, and (b) shows the CDF of all similarities between summaries.

3 Measurement Study on Gaze Patterns

3.1 Collection Experiment Design

In order to ensure the quality of manual summaries, we recruited 50 college students whose mother tongue was Chinese to participate in the collection experiment. We used a corpus of 100 Chinese news articles in 10 categories to collect gaze behavior. Three *Tobii Eye Tracking 4C* devices were used to track the gaze behavior of different users when they were reading given news articles. The eye-tracking device samples a gaze point every 10 milliseconds or so. Each article is between 200 and 800 Chinese characters long. After reading a given article, the user was asked to refer to the original content and the title of the article to provide a summary that meets his or her own preference. In addition, there are no restrictions on the reading time or the summary content.

3.2 Overview of Dataset

A total of 4998 samples were collected, including 3498 samples in the training set, 500 samples in the validation set, and 1000 samples in the test set.

We calculated the cosine similarity between the summaries given by different users for the same article. The average similarity is 0.383. And similarities between each user and others are shown in Fig. 1(a). As can be seen from the figure, there are low similarities between different users. In particular, user 8, who prefers to use many words that don't exist in original articles, has the lowest similarity value of 0.123. From the CDF in Fig. 1(b), it's obvious that more than 70% of all similarity values do not exceed 0.5.

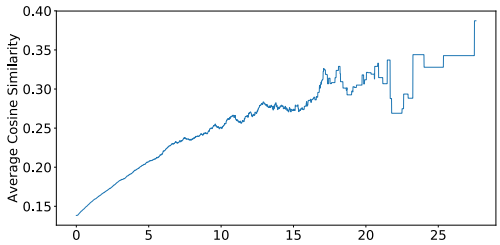
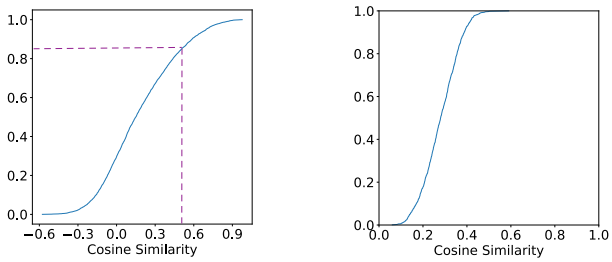


Figure 2: The average cosine similarity of sentences and summaries at different gaze times on sentences.



(a) (b)

Figure 3: (a) shows the CDF of all similarities between the POS distribution of gaze time and that of words that constituting the corresponding summary, and (b) is the CDF of all similarities between the average POS distributions of gaze time of different users.

3.3 Gaze Patterns

3.3.1 Distribution of gaze time on sentences

We counted the gaze time of each person reading each sentence and calculated the cosine similarity between the current sentence and the summary given by the user. Then, the similarities at the same gaze time are averaged to obtain Fig. 2.

It can be seen in Fig. 2 that the longer the gaze time on the sentence, the higher the similarity between the sentence and the summary given by the user. However, in the actual application scenario, we can't collect the user's gaze behavior in real time. Therefore, the result presented in Fig. 2 cannot be used directly. However, we believe that the result at the sentence level can be mapped to some intrinsic property of the sentence, which can be used in the personalized text summarization model.

3.3.2 Distribution of gaze time on POSs

POS is an intrinsic property of articles and sentences. We calculated the POS distribution of gaze time and

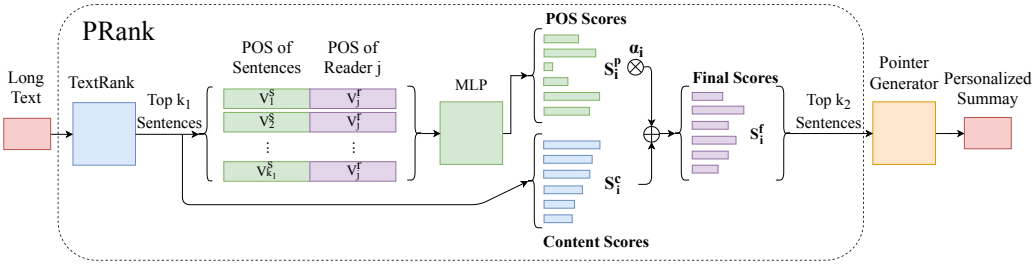


Figure 4: Our model with PRank and pointer generator for personalized text summarization

that of words constituting the corresponding summary, and the cosine similarity between them. Since the proportions of different POSs in the natural corpus are different, in order to more clearly reveal the impact of each POS in the final result, we normalize each distribution before calculating the final similarity. The CDF of all the calculated similarities is shown in Fig. 3(a). As can be seen from it, around 15.4% of the values are higher than 0.5. After that, we calculated the average POS distribution of gaze time of each user. The CDF of the similarities between the average POS distributions of different users is shown in Fig. 3(b). Obviously, the similarities between different users are very low. Almost all the values are less than 0.5.

4 Neural Architecture of Personalized Text Summarization

The model we proposed is an integration of an extraction module named PRank and a mature abstraction module named pointer generator. And We will describe PRank in detail and briefly introduce pointer generator in the following parts. The overall frame diagram of our model is shown in Fig. 4.

4.1 PRank

PRank takes into account both the content of sentences and the user's preferences. The whole process is as follows:

- *Content Score (CS) Calculation:* We first use TextRank to select k_1 sentences with higher scores. TextRank is an algorithm similar to PageRank, which is more concerned with the importance of sentence content throughout the text. The content score for each selected sentence is marked as S_i^c.
- *POS Score (PS) Calculation:* For each sentence selected in the previous step, the corresponding POS distribution vector V_i^s of words can be obtained statistically. For current user j, we can extract the average POS distribution vector V_j^r from the dataset as his personalized POS preference. And then V_i^s and V_j^r are fed into the following MLP to calculate the POS score S_i^p for each sentence.
- *Final Score Calculation:* The final score of each sentence is calculated as follows:

$$S_i^f = S_i^c + \alpha_i \cdot S_i^p \tag{1}$$

α_i is a learnable parameter. The k_2 sentences with the highest final scores are then fed to pointer generator to generate a personalized summary.

Through the above process, we can guarantee that the final summary can not only be related to the main idea of the article but also reflect the user's preference.

4.2 Pointer Generator based on PRank

Pointer generator is a hybrid between a sequence-to-sequence model with attention mechanism and a pointer network [14] which is effective in solving the problem of OOV words. Besides, it includes a coverage mechanism that addresses the problem of duplicate words.

For the same text, the sentences extracted by PRank are different due to different POS preferences so that pointer generator can generate different personalized summaries for different users. In the actual process, due to the limited amount of text with gaze records, it is necessary to pre-train pointer generator using a same type of dataset, and then use our dataset to finetune it. Our dataset is also used to train PRank. We define the PRank loss function for the input text corresponding to the article m and the user n:

$$loss_{mn} = -\log \sum_{i_{selected}} [S_i^f \cdot ROUGE(g_{mn}, r_{mn})] \tag{2}$$

S_i^f is the final score calculated by PRank. g_{mn} and r_{mn} represent the generated summary and the reference summary respectively.

Models	ROUGE		
	1	2	L
Lead-1	25.1	9.8	17.7
Lead-2	22.6	8.5	13.9
TextRank-1	20.3	6.9	13.1
TextRank-2	17.4	5.7	10.0
PGN	38.4	18.8	33.5
TextRank+PGN	37.2	17.9	32.2
PRank+PGN(Ours)	**41.5**	**21.4**	**36.8**

Table 1: The performances on our dataset.

5 Evaluation

5.1 Dataset

- The TTNews corpus is a large Chinese single-document summary corpus that is used in the experiment to pre-train pointer generator.
- Our dataset is divided into three parts, a training set, a validation set, and a test set. Before training the PRank module, we need to finetune pointer generator.

5.2 Comparison models

- *Lead-1,2:* Select the first one or two sentences as the summary.
- *TextRank-1,2:* Select the top one or two sentences with higher scores calculated by TextRank as the summary.
- *Pointer generator(PGN):* Feed the entire original article into the model to generate a summary.
- *TextRank+PGN:* The k_2 sentences extracted by TextRank are fed into PGN to generate a summary.

5.3 Experiment Setting

In PRank module, we select 25 classes of POS to form the POS vector both for each sentence and each user. The MLP model has one input layer, three hidden layers, and one output layer. And the corresponding hidden sizes are set to 256, 128, 256 and 128. Considering the average length of the articles, k_1 is set to 6, and k_2 is set to 3.

Models	ROUGE		
	1	2	L
PGN	38.4	18.8	33.5
PRank(Only CS)+PGN	37.2	17.9	32.2
PRank(Only PS)+PGN	40.8	20.7	35.9
PRank+PGN	**41.5**	**21.4**	**36.8**

Table 2: Comparison of different model configurations on our dataset.

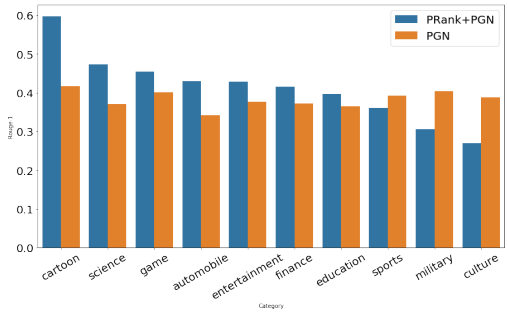


Figure 5: Comparison on different categories.

5.4 Results and Analyses

The specific comparison results on the test set are shown in Table 1. We evaluate our model and baseline models with the standard ROUGE metric.

The same input text corresponds to many different reference summaries given by different users, and each user is concerned with different parts of the original text. Therefore, pointer generator that takes the entire articles as input generates more satisfying summaries for many more users than the model, which only uses the fixed several sentences selected by TextRank. This is why the ROUGE score for pointer generator exceeds the model composed of TextRank and pointer generator.

However, our proposed novel extraction method PRank can select different k_2 sentences according to the preferences of different users. And more selective input of partial text instead of the whole text can maximize the performance of the following abstractive model. So as we can see from Table 1, our model outperforms the best baseline model by 3.1% on ROUGE-1, 2.6% on ROUGE-2 and 3.3% on ROUGE-L. For illustration, Table 3 shows some results of our model.

In addition, we also carried out an experimental analysis of the performance of different parts of the proposed model. Table 2 shows the ROUGE for different model configurations. It can be seen that the gain of the POS score to our model is much greater than that of the content score, and the performance of the

Table 3: Sample summaries generated by our model

Original Text: 网易娱乐 9 月 8 日报道北京时间 9 月 8 日消息，据香港媒体报导，六场《金光灿烂徐小凤演唱会 2018》昨晚在红馆揭开序幕， 全场爆满 。... 一向幽默风趣的她还现场 说冷笑话 ，当她表示 三年不见 以为歌迷不认得她，... 徐小凤 特别选唱陈百强 （Danny）的《有了你》 怀念故友 ，她忆述：“当年在片场候场时，他把我拉到一边弹琴我听，虽然不知道他弹什么，应该是正在弹一些新作品；送给所有喜欢 Danny 歌曲的朋友，包括我在内。”

Netease Entertainment reported on September 8th, Beijing time on September 8th news, according to Hong Kong media reports, six "Golden Brilliant Xiaofeng Xu Concert 2018" kicked off in the Red Pavilion last night where was full of the audience . ... She, who is always humorous; also said a joke on the scene when she said that she thought she wasn't recognized by the fans because of not appearing for three years , ... Xiaofeng Xu specially sang Danny Chen's (Danny) "With You" to miss her old friend , and she recalled: "When I was on the studio, he pulled me aside and played the piano for me. Although I didn't know what he was playing, he should be playing some new works; for all the friends who liked Danny's songs, including me."

Reference Summary(User 20): 徐小凤再开演唱会，全场爆满唱金曲说金句，特别怀念故友陈百强

Xiaofeng Xu opened the concert again where was full of the audience, singing golden songs, saying golden sentences and specially missing her old friend, Danny Chen .

Generated Summary(User 20): 组图：徐小凤 特选唱陈百强怀念故友 ，有点儿不好意思说：“你真的对我唱歌多年”（图）

Photo: Xiaofeng Xu specially sang Danny Chen's to miss her old friend , and was a little embarrassed to say: "You really sang to me for many years" (Figure)

Reference Summary(User 44): 时隔三年 ，徐小凤出现演唱会 说冷笑话 与粉丝互动

After three years , Xiaofeng Xu appeared in a concert, saying cold jokes and interacting with fans.

Generated Summary(User 44):[拌饭] 徐小凤特别选唱陈百强；幽默风趣：还现场 说冷笑话 (图)；三年不见 以为歌迷不认得。

[Bibimbap] Xiaofeng Xu specially sang Danny Chen's; humorous and funny: also said a cold joke on the scene (picture); she thought she wasn't recognized by the fans because of not appearing for three years .

Reference Summary(User 47): 徐小凤演唱会首日 爆满

Xiaofeng Xu' s concert was full of the audience on the first day

Generated Summary(User 47): 组图：六场《金光灿烂徐小凤演唱会 2018》昨晚在红馆揭开序幕， 全场爆满 。

Set of pictures: six "Golden Brilliant Xiaofeng Xu Concert 2018" kicked off in the Red Pavilion last night where was full of the audience .

model only retaining the content score is even lower than that of PGN without PRank.

As can be seen from Table 3, the three different users in the example have different emphasis on the original text. The contents in different color blocks represent the unique concerns of different users corresponding to which is marked in the same color in the original text. At the same time, our model generates a more personalized and appropriate summary based on the unique focus of different users.

Furthermore, we selected the best performing baseline to compare with our model. Our dataset contains a total of ten categories of articles. It is shown in Fig. 5 that our model performs better than the baseline in most categories of articles. And our model outperforms the baseline by 18 ROUGE-1 points for cartoon articles. We also noticed that for military, sports, and cultural articles, the performance of our model is worse than that of the baseline. The result inspires us that the type of article should also be considered.

6 Conclusion

In this paper, we propose a model integrating a gaze-based key sentence extracting strategy, PRank, and pointer generator for personalized text summarization. The results show that our model significantly outperforms other baselines.

However, in this study, we only used gaze features in the sentence extraction phase, which led to the theoretical upper limit of the performance of our model. In our future work, we'll incorporate more features into an abstract generator model for better performance.

Acknowledgement

This work is supported in part by NSFC under Grant No.~61872215, SZSTI under Grant No.~JCYJ20180306174057899, and Shenzhen Nanshan District Ling-Hang Team Grant under No. LHTD20170005.

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